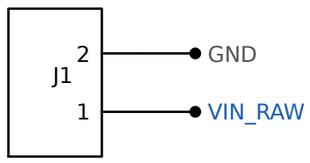
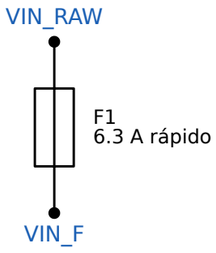
Cubo AP · Placa de control v1 — esquema

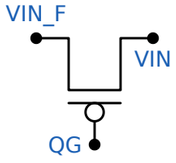
Generado de diseno.py (la misma lista de conexiones que la placa). Las etiquetas azules son nombres de red: pines con la misma etiqueta están conectados. Sin probar en hardware.

1 · Entrada del cartucho y protección

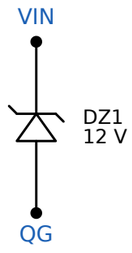


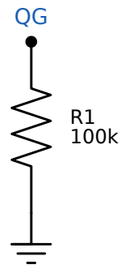
Entrada 12–36 V

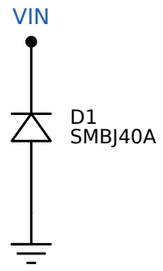


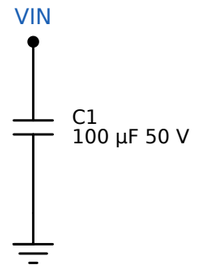


Q1 · P-MOSFET



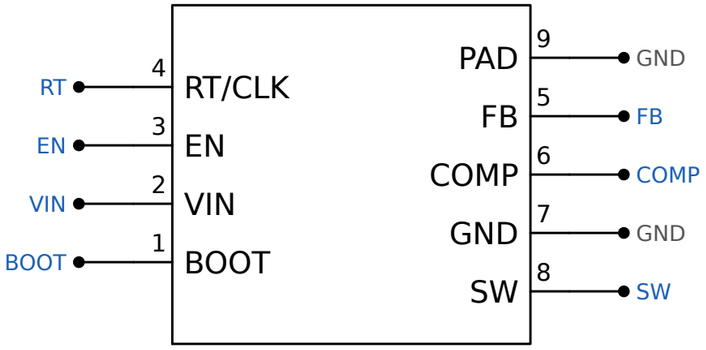




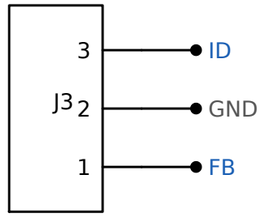


TP1: VIN

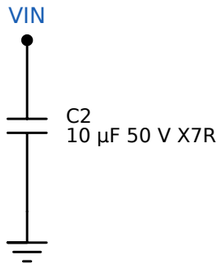
Q1 protege contra inversión de polaridad (el diodo interno conduce y luego el canal se enciende). DZ1 limita Vgs a 12 V. D1 recorta transitorios del cartucho.

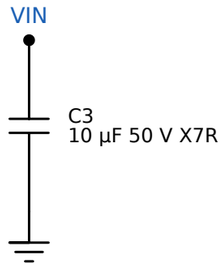


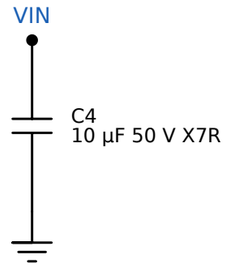
U1 · TPS54560DDA

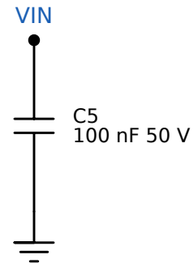


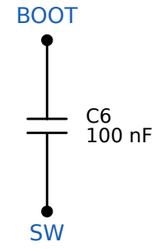
Tarjeta

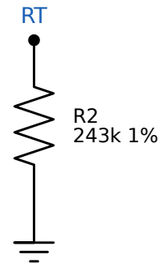


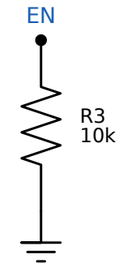


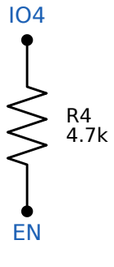


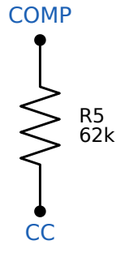


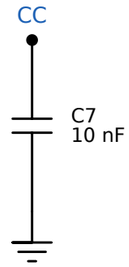


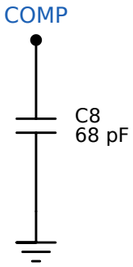


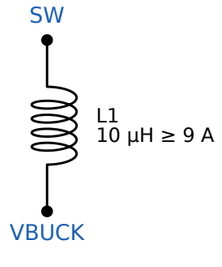


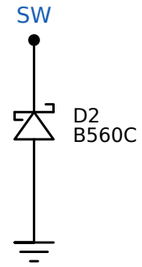


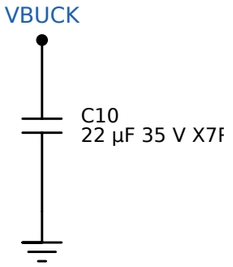


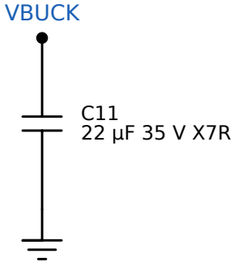


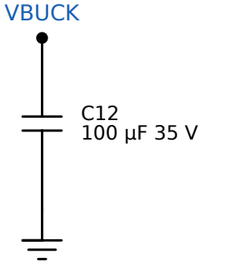


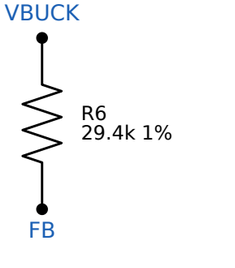


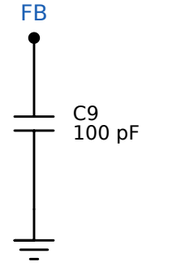


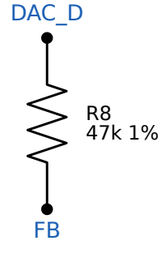


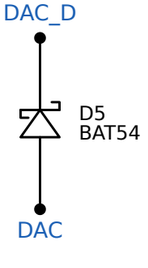


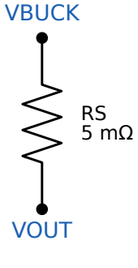


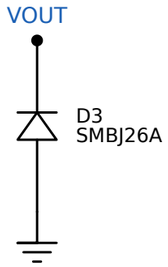








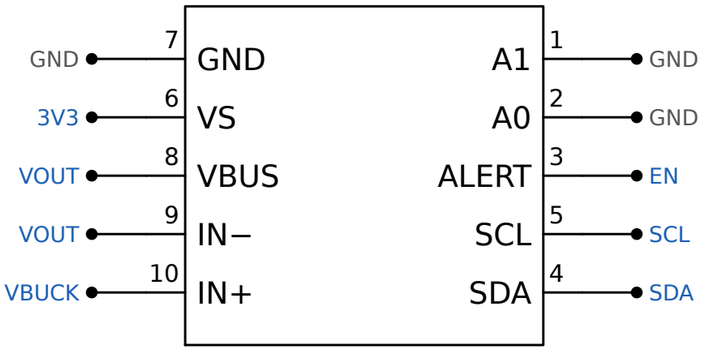




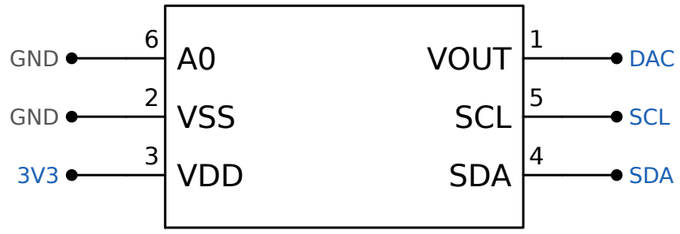
TP4: VOUT

Vout = 0.8 V × (1 + 29.4k / R_B). R_B vive en la tarjeta (J3 pin 1–2): sin tarjeta, FB = Vout y la salida se cae a ~0.8 V. El DAC solo inyecta corriente al FB a través de D5: solo puede BAJAR el voltaje. Compensación tipo II para cruce ≈ 15 kHz a 12 V (verificar con la herramienta de TI). RS = 5 mΩ, sentido Kelvin al INA226.

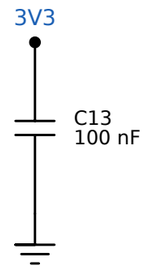
3 · Medición y ajuste: INA226, DAC MCP4725, ID, Vin y temperatura

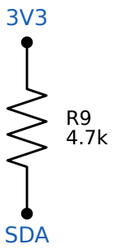


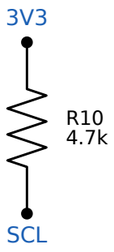
U4 · INA226

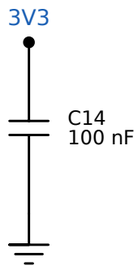


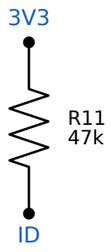
U5 · MCP4725A0

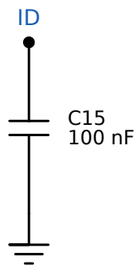


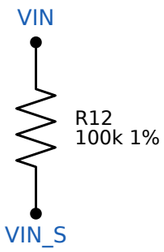


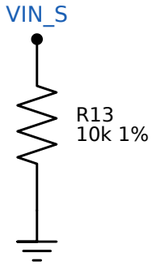


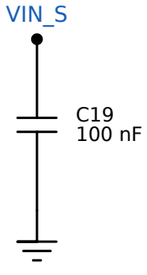


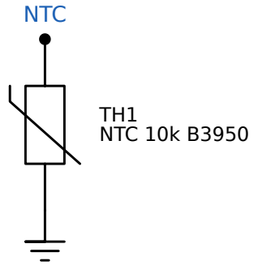


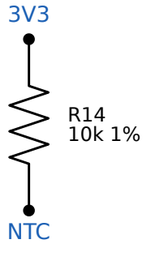






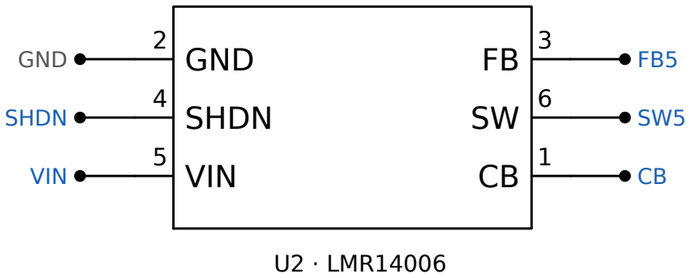




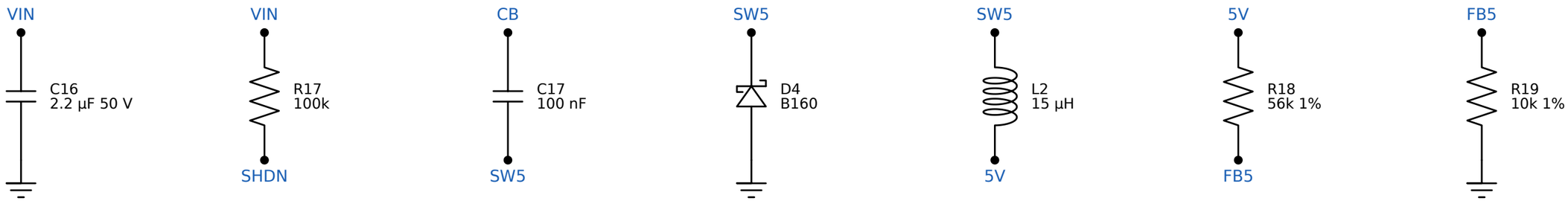


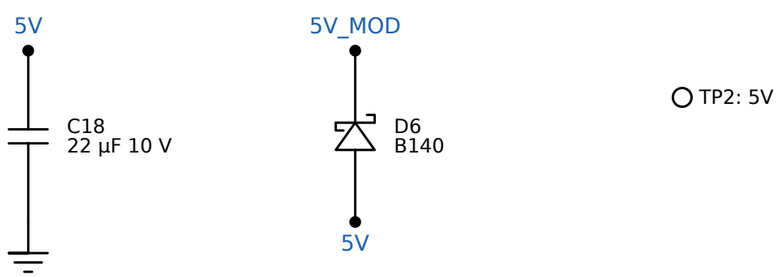
INA226 en 0x40 (A0 = A1 = GND), DAC en 0x60. ALERT del INA226 está en la red EN: al dispararse tira el EN del reductor a GND por hardware (~1 ms). Pull-up de ID = 47k (R_ID de la tarjeta: 10k…150k).

4 · 5 V auxiliar (LMR14006)



U2 · LMR14006

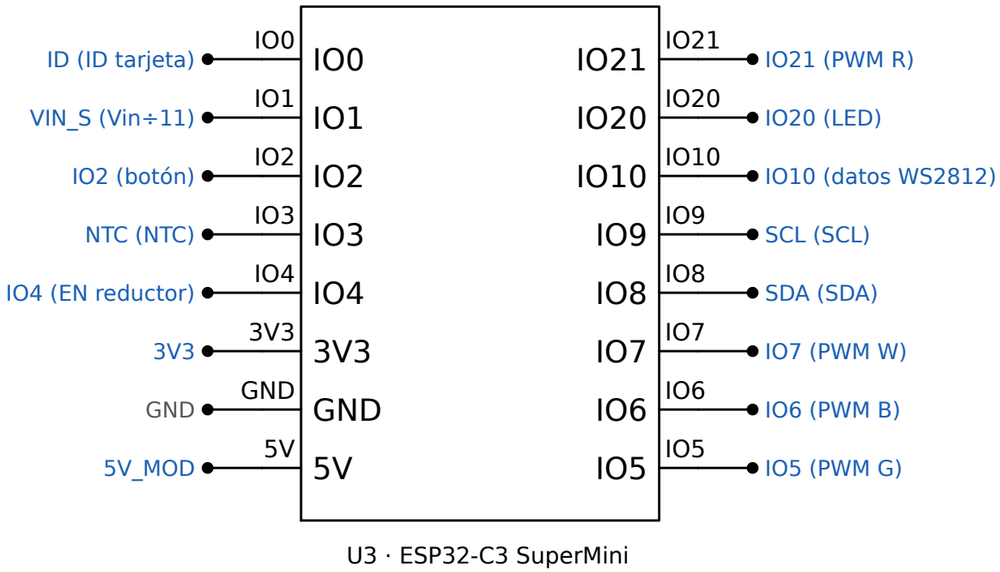




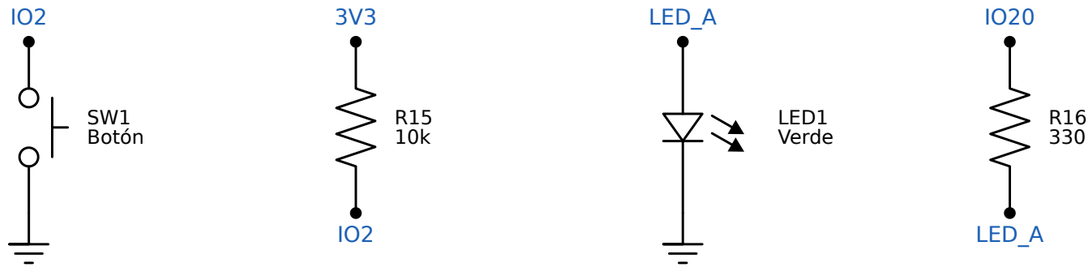
TP2: 5V

5 V para drivers, buffer WS2812 y el SuperMini. D6 evita alimentar al USB de la PC cuando se programa con la placa conectada.

5 · ESP32-C3, botón y LED de estado

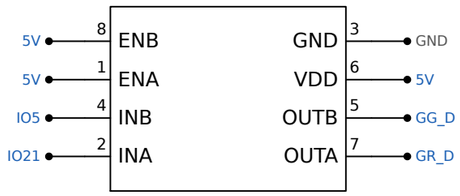


U3 · ESP32-C3 SuperMini

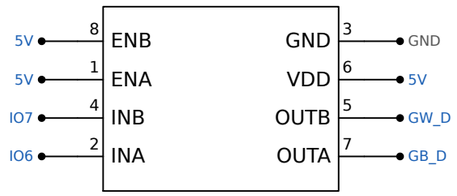


El SuperMini va en tira hembra (enchufable). Pines elegidos: ADC (IO0–IO3) a la izquierda, PWM/I2C/datos a la derecha. IO4 = EN del reductor (R3 lo mantiene apagado al arrancar).

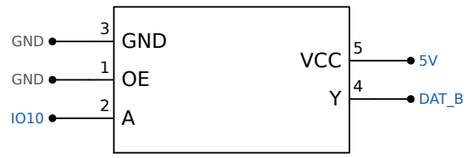
6 · Canales PWM (UCC27524 + MOSFET), datos WS2812 y salida



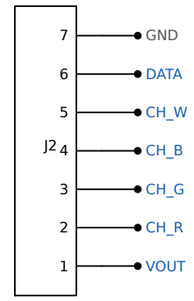
U6 · UCC27524



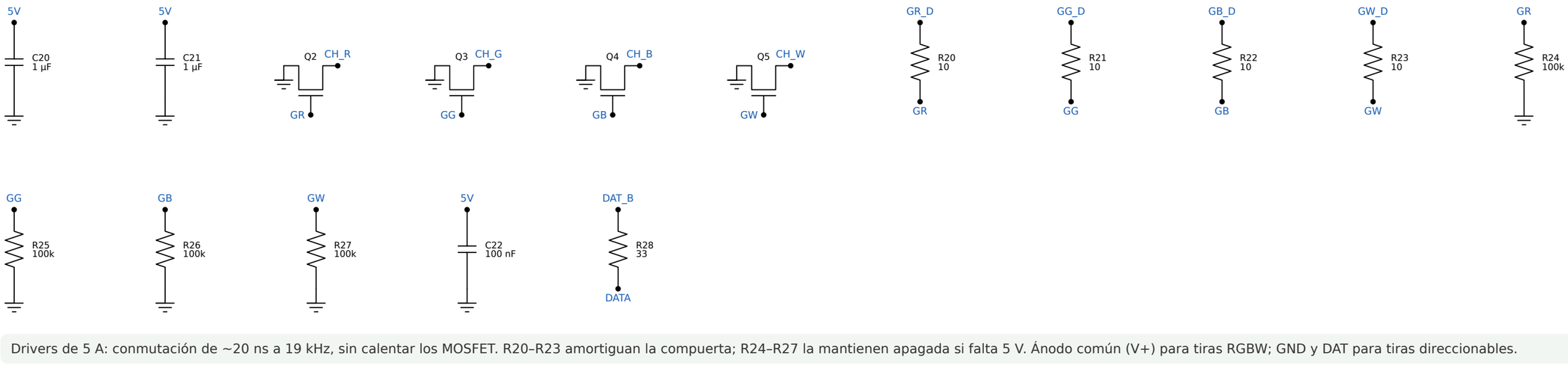
U7 · UCC27524



U8 · 74AHCT1G125



Salida LED



Drivers de 5 A: conmutación de ~20 ns a 19 kHz, sin calentar los MOSFET. R20–R23 amortiguan la compuerta; R24–R27 la mantienen apagada si falta 5 V. Ánodo común (V+) para tiras RGBW; GND y DAT para tiras direccionables.